

Who Opens Their Homes to Refugees? Social ties and capacity during the Ukrainian refugee crisis

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Abstract

Private hosting of refugees is a high-cost prosocial behaviour: households share their homes for months. Whether it can scale during displacement crises depends on who steps forward, but population-level evidence is scarce because hosts are rarely identifiable in the full population. Using Denmark's linked address, population and housing registers, we identify about 3,900 households that privately hosted Ukrainian refugees after Russia's 2022 invasion and compare them to all other households. Hosting is highly selective along two margins. Ukrainian-linked households are far more likely to host, and workplace exposure to Ukrainians is associated with hosting. Hosts also come disproportionately from larger households in spacious homes. Selection on socioeconomic resources, including owner-occupation, is concentrated among non-Ukrainian hosts, whereas Ukrainian-linked households host across the income distribution. The findings clarify when private hosting can provide flexible civic reception capacity during displacement shocks.

Sudden displacement crises generate intense demands on host-society institutions and on individual citizens. Within five weeks of Russia’s February 2022 full-scale invasion, more than 4 million people had fled Ukraine and needed accommodation across Europe. The scale of that shock triggered an extraordinary response from private households. Private hosting emerged alongside ad hoc public shelters as a major mode of reception: residents opened their homes to newly arrived refugees, often people they did not previously know, contributing scarce living space, material support, and local knowledge over months (Haller et al. 2022; Bassoli and Luccioni 2024; Carlsen, Gårdhus, and Toubøl 2024; Herpell et al. 2025). Survey-based estimates suggest that roughly 27% of Ukrainian refugees in Europe were privately hosted in the first months after the invasion (European Union Agency for Asylum, International Organization for Migration, and Organisation for Economic Co-operation and Development 2022). This made private hosting more than an emergency source of beds. By placing refugees inside ordinary households, it could provide a more personal form of initial reception, with access to local routines, practical information, and social contacts that are harder to supply in impersonal shelter settings.

At the same time, private hosting is a particularly demanding form of prosocial behaviour. Unlike low-cost donations or signed petitions, it requires households to share their dwelling with people who in many cases were outside their close pre-crisis circle and may have been complete strangers or connected only through weak ties, absorbing both material and psychological costs. It is a revealed high-cost choice rather than a self-reported willingness to help, and it scaled rapidly during a crisis that overwhelmed public reception systems (Boin et al. 2005). Displacement crises are often temporally uneven: accommodation demand can spike within days or weeks and then recede as arrivals slow or refugees move into longer-term housing. From a reception-system perspective, this makes flexibility valuable: private hosting can expand quickly during arrival surges and contract as needs recede, avoiding some of the fixed costs of maintaining public shelter capacity under uncertain demand. But it is not free capacity; it shifts material, coordination, and emotional costs onto households, making the distribution of household resources and social ties central to whether it can be mobilised at scale.

Whether such revealed solidarity can scale during future displacement crises depends on who steps forward. Existing research on private hosting has documented host motivations, host activities, and the scale-up of hosting schemes (Bassoli and Luccioni 2024; Monforte, Maestri, and d’Halluin 2021; Haller et al. 2022; Tomlinson et al. 2023). On refugee outcomes, Herpell et al. (2025) show that private hosting improved the integration prospects of Ukrainian refugees in Germany, but for users of a single matching

platform; in companion work on Denmark, Foged et al. (2026) estimate the consequences of initial accommodation type using the same administrative registers as those used here. What is missing is population-level evidence on who, in the underlying population, actually steps forward—and how the resources and ties that enable participation are distributed across households. This gap is partly empirical: private hosts are rarely identifiable in full-population data because hosting is usually not recorded as a dedicated administrative status. As a result, most studies can observe refugees, platform users, or volunteers who opt into a survey, but not realised hosts against the full set of households that could potentially have hosted. Understanding the structural foundations of costly civic provision is essential for assessing when private hosting can supplement, or partially substitute for, public reception systems (Ostrom 1996; Feischmidt, Pries, and Cantat 2019).

Two broad mechanisms could plausibly structure who hosts. The first is *capacity*: hosting requires room to accommodate additional people and the financial resilience to absorb day-to-day costs, and these resources are unevenly distributed across the population (Andreoni 1990; Piliavin and Charng 1990; Wilson and Musick 1997). Experimental work on refugee helping likewise shows that willingness to help falls as personal costs rise (Böhm et al. 2018). The second is *social proximity*: shared language, prior migration experience, or workplace exposure to Ukrainians can lower information and coordination costs, reduce uncertainty, and create stronger moral obligations to help (Granovetter 1973; Putnam 2000; Portes 1998; Cheung and Phillimore 2014). This emphasis on connection is also consistent with evidence that shared identity and perceived closeness increase helping toward refugees and other socially distant others (Levine et al. 2005; Hellmann, Fiedler, and Glöckner 2021). These mechanisms generate distinct empirical predictions. If capacity is the binding constraint, hosting should concentrate among households with slack housing and stable resources, especially those without prior Ukrainian ties. If social proximity is central, households with Ukrainian links should host much more frequently, and conditional on those ties additional selection on income or housing quality may be weaker. Equally informative is what may turn out to be *less* predictive than one might expect: if hosting is mainly structured by room and ties, then generic markers of advantage or broad local context should add less once those two margins are taken into account.

To evaluate these mechanisms empirically, we shift attention to the complementary supply side. Using Denmark’s full-population administrative registers, we identify the roughly 3,900 households that privately hosted Ukrainian refugees following Russia’s full-scale invasion and compare them to all other households in the country. This is an unusual measurement setting. Linked address histories allow us to recover realised

hosts from the full population rather than from a self-selected sample of volunteers or platform users. Crucially, those person-level records can also be linked to Denmark's housing registry, allowing us to observe dwelling characteristics for every household, including living space, rooms, tenure, and housing type. We can therefore study not only who hosted, but also the material housing capacity that made hosting feasible. More broadly, the data allow us to observe socio-demographic characteristics, household composition, housing conditions, labour-market position, migration background, and prior exposure to Ukrainian networks for the entire Danish population.

We document selection patterns descriptively, estimate parametric and non-parametric prediction models, and split the analysis by whether a Ukrainian citizen was already present in the household to test the capacity-versus-proximity contrast. The motivation for this split is that the reasons for hosting are likely to differ across households with and without a direct pre-existing Ukrainian household tie. We then examine how the same factors are associated with the duration of hosting arrangements, since whether costly helping is sustained matters as much as whether it is initiated. The results indicate that costly hosting is concentrated in households combining housing capacity and social ties, that the relative weight of these two mechanisms differs sharply across population subgroups, and that hosting arrangements linked to prior ties appear more durable. We discuss what these structural conditions imply for when civic provision can supplement public reception systems during future displacement shocks.

Results

Scale of the inflow and the hosting population

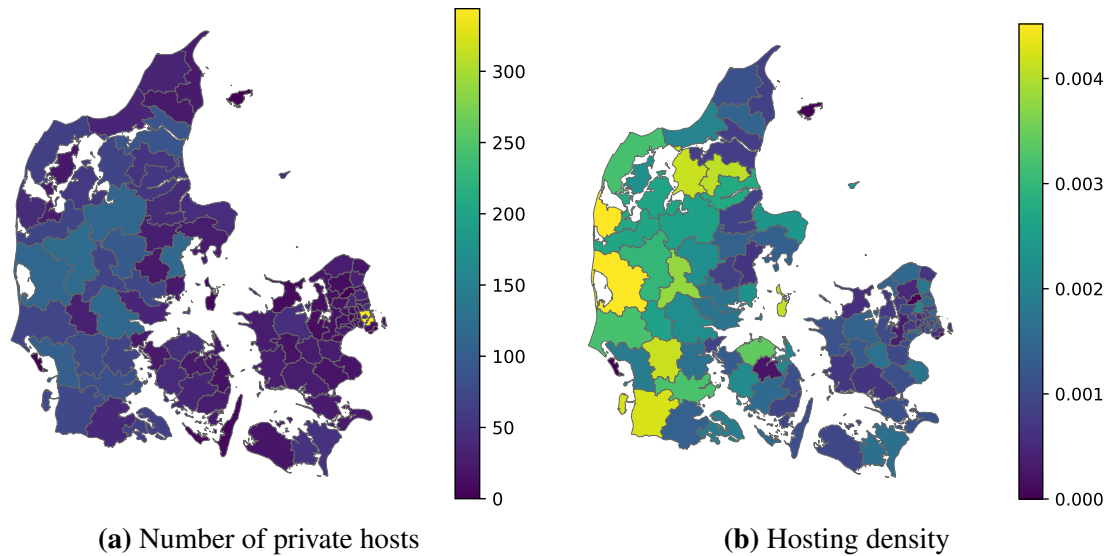
Between March and June 2022, approximately 25,000 Ukrainian refugees arrived in Denmark; the cumulative total reached roughly 40,000 by the end of 2023, a tenfold increase over baseline flows. Public reception capacity could not absorb the inflow during the peak weeks, and private households accommodated a substantial share of arrivals: roughly 41% of refugees arriving during the peak were privately hosted, with the remainder placed in conventional public refugee housing or in ad hoc pop-up shelters (Foged et al. 2026). Out of the roughly 2.67 million Danish households we observe in the registers, about 3,900 (0.15%) privately hosted Ukrainians during this period. Together, these two figures point to a first basic lesson: even when only a small fraction of households steps forward, private hosting can absorb a substantial share of a sudden inflow. That hosting is rare at the population level is consistent with its costliness—

sharing a dwelling with newly arrived refugees for months is a much steeper ask than a donation or a signed petition. The remainder of the analysis compares these hosting households to the rest of the Danish household population to ask who, in the underlying population, actually does it.

Geographic distribution and household-level selection

Private hosting of Ukrainian refugees was geographically widespread but unevenly distributed across Denmark. Figure 1 shows substantial variation across municipalities in both the absolute number of hosting households and hosting density. The raw count is naturally highest in the largest municipalities, but the density map indicates that hosting was not exclusively an urban phenomenon: it also emerged strongly in smaller municipalities and rural areas.

Figure 1: Private hosting of Ukrainian refugees across Danish municipalities.



Note: Panel (a) shows the raw number of hosting households per municipality. Panel (b) shows hosting density, measured as the number of private hosts divided by the total household count in each municipality.

At the household level, the contrasts between hosts and the rest of the Danish population are sharp. Supplementary Table S1 compares hosting households to the rest of Denmark, with person-level characteristics measured for each household's breadwinner; we report these comparisons both pooled and split by whether the household included a Ukrainian member at the end of 2021. It is worth emphasising that Denmark had no large Ukrainian diaspora before the crisis: only about 0.3% of Danish residents were Ukrainian citizens at the end of 2021, and only about 6,000 Danish households included a Ukrainian member at the time. Even so, hosting rates differ sharply across the

two groups: among Danish households without a Ukrainian member at end-2021, only about 0.07% hosted; among households with at least one Ukrainian member, roughly 31% did—a more than 400-fold difference before any other characteristic is taken into account, foreshadowing the central role of social proximity in what follows. Ukrainian household membership is thus a low-prevalence, high-impact predictor: rare in the population, but enormously informative conditional on its presence. This shapes how we interpret coefficients and importance scores throughout, and is why we report rescaled rather than raw effects. It also motivates analysing households with and without prior Ukrainian ties separately, since pooling the two groups could obscure distinct selection patterns. Beyond this gap, hosting households have breadwinners who are younger, somewhat more likely to be male and married, more highly educated, and more strongly attached to the labour market than the average Danish household. Hosting households are also much larger and disproportionately concentrated in single-family homes, pointing immediately to the importance of both space and household structure. Hosts are also much less likely to belong to the Danish state church than non-hosts, a gap that is only partly explained by the presence of Ukrainian-linked households and that persists conditional on other characteristics in the regression models.

The split by pre-existing Ukrainian household ties is especially revealing. Among households without a Ukrainian member, hosts are strongly positively selected relative to other households without a Ukrainian member: they are richer, better educated, and much more likely to live in spacious dwellings. Some occupational differences also appear, but the dominant contrasts are in socioeconomic resources and housing capacity. Many realised hosting arrangements therefore formed outside a pre-existing co-resident Ukrainian tie, consistent with a large margin of stranger or weak-tie helping, even if our data do not allow us to distinguish cleanly between complete strangers and lighter forms of prior connection. Among households with a Ukrainian member already present, by contrast, many of these differences attenuate sharply. Hosting among Ukrainian-linked households is much less concentrated among the most socioeconomically advantaged households, consistent with a pattern in which pre-existing ties lower matching frictions, reduce the social and practical costs of co-residence, and strengthen perceived obligations to help, allowing households to step forward without the same degree of exceptional resource slack that appears more important elsewhere.

Income and housing capacity

The same contrast is visible when private hosts are compared within their own subgroup rather than against a single pooled national baseline. Among households without

a Ukrainian member, Figure 2(a) shows that hosts are heavily overrepresented in the upper tail of that subgroup's income distribution. Private hosting in this group was therefore not a generic middle-of-the-distribution response; it came disproportionately from relatively advantaged households. Among Ukrainian-linked households, by contrast, hosts track their subgroup's own income distribution much more closely, indicating that hosting is comparatively uniform within this group even though Ukrainian-linked households as a whole are concentrated toward the lower part of the national income distribution.

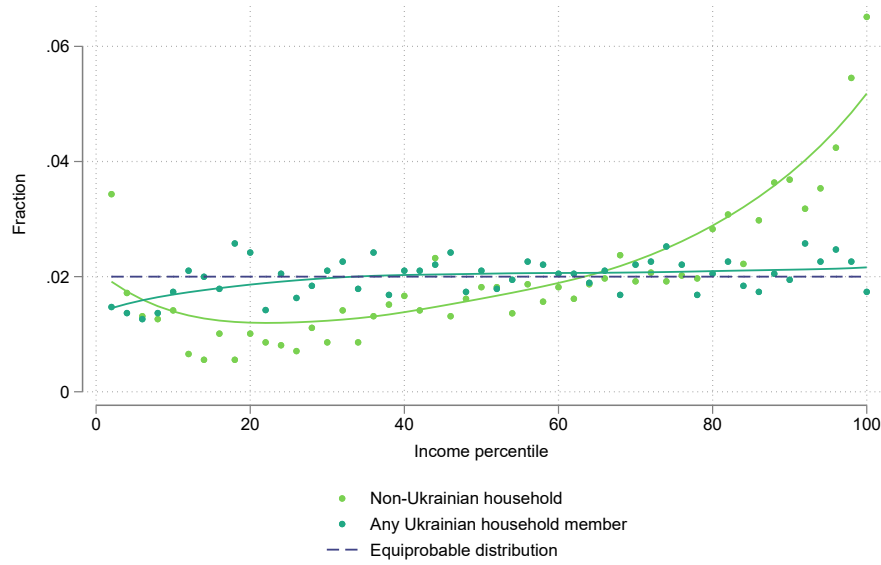
Housing capacity displays the same subgroup-conditional pattern, more starkly. Among households without a Ukrainian member, Figure 2(b) shows that hosts are strongly concentrated in the upper tail of that subgroup's per-capita living-space distribution, indicating that substantial slack housing capacity was especially important in this group. Among Ukrainian-linked households, by contrast, hosts again track their subgroup's own distribution much more closely, even though Ukrainian-linked households as a whole are concentrated toward the lower part of the national living-space distribution. Willingness to host alone is insufficient: households must also have the room to do so.

Joint predictors of hosting

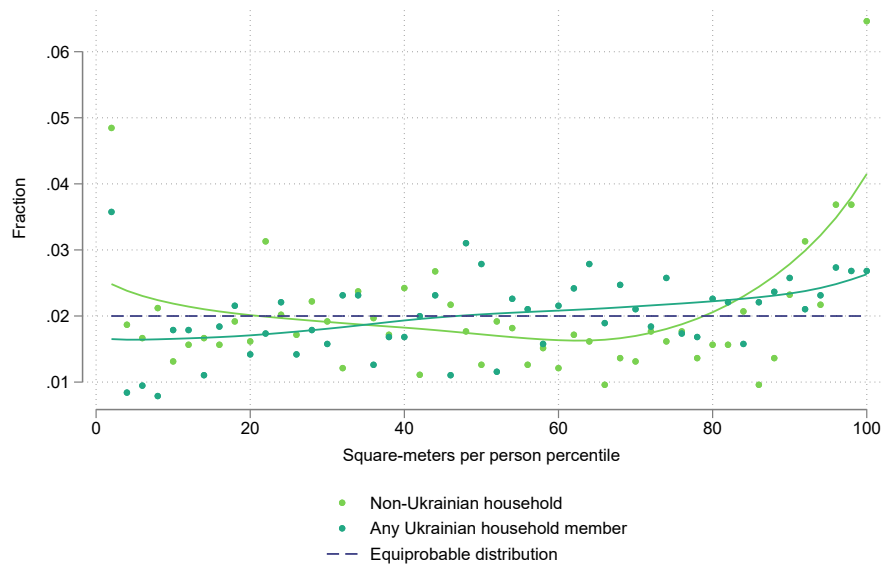
Many of the relevant predictors are correlated. We therefore turn to multivariate models that ask which factors remain informative when considered jointly. Supplementary Table S2 reports linear probability models for the full population and for the two subsamples defined by pre-existing Ukrainian household ties; municipality fixed effects absorb broad local differences in housing markets, refugee inflows, and policy environments. Figure 3 reports the corresponding coefficients rescaled by group-specific baseline hosting prevalence, allowing them to be read as proportional changes relative to the relevant comparison group's hosting rate.

Three patterns survive the multivariate specification. First, household ties to Ukrainians are the single strongest predictor of hosting: a household that already includes a Ukrainian member is far more likely to host, even after controlling jointly for a wide range of other characteristics (Supplementary Table S2). Second, housing capacity remains central: square metres per person, rooms per person, and household size continue to predict hosting strongly. Third, workplace exposure also matters: breadwinners employed in workplaces with a high Ukrainian share are more likely to host, suggesting that social proximity outside the household also lowered the barriers to participation. Equally notable, several intuitive predictors are much weaker once these margins are accounted for. In the non-FE specifications, municipality-level covariates are modest

Figure 2: Distribution of private hosts across the income and living-space distributions.



(a) Income distribution



(b) Living-space distribution

Note: Panel (a) shows the distribution of breadwinners in private-hosting households across the income distribution of their respective comparison population (households without or with a Ukrainian member); Panel (b) shows the corresponding distribution across square metres per person. Each point corresponds to a two-percentile bin, benchmarked against the relevant comparison population. The horizontal line labelled “equiprobable distribution” indicates the distribution that would arise if private hosts mirrored the distribution of the comparison population. The concentration of hosts without a Ukrainian household member in the upper tail of both distributions indicates that socioeconomic resources and slack housing capacity were major margins of selection into hosting.

Figure 3: Predictors of selection into private hosting, rescaled by baseline prevalence.



Note: Coefficients from linear probability models for selection into private hosting, rescaled by the mean of the dependent variable in the relevant subgroup. Rescaled coefficients can be interpreted as proportional changes in the probability of hosting relative to baseline prevalence. The two splits by pre-existing Ukrainian household ties are reported side by side. Underlying unscaled estimates appear in Supplementary Table S2.

predictors relative to household ties and housing capacity; in the FE specifications, these local-context measures are absorbed by municipality fixed effects and are therefore not separately estimated. Among Ukrainian-linked households, many of the usual markers of socioeconomic advantage are far less predictive than one might expect if hosting were driven mainly by general affluence (Supplementary Table S2, columns 5–6; Figure 3).

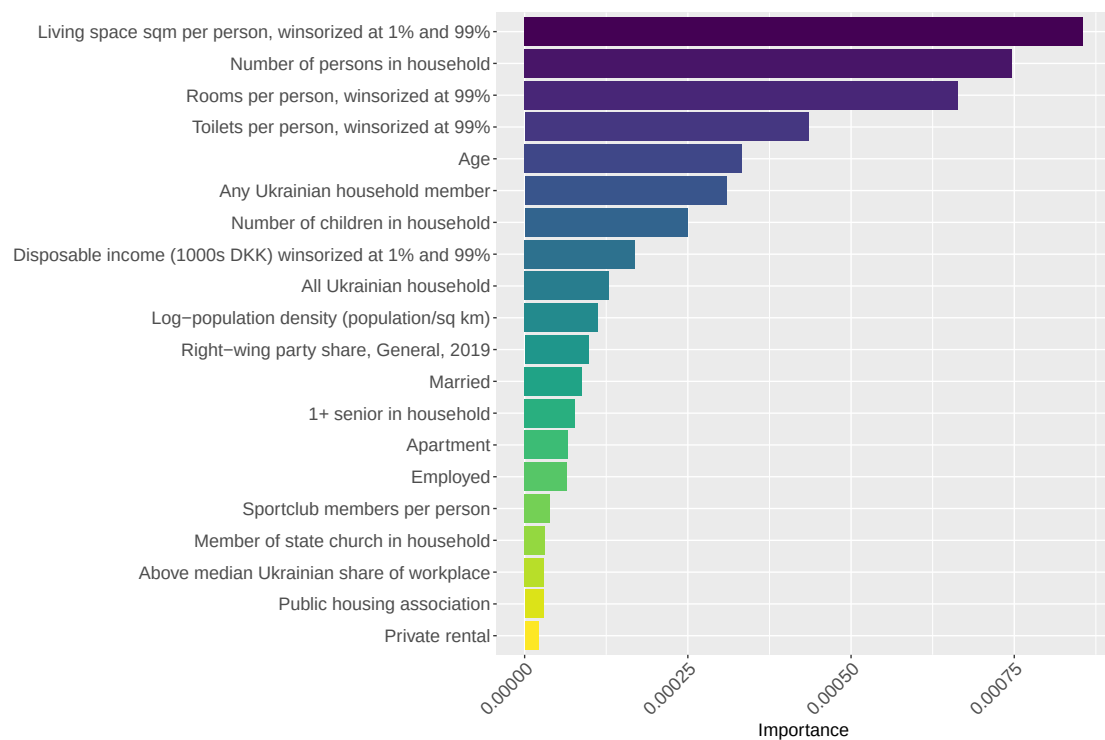
The split sample sharpens the interpretation. Among households without a Ukrainian member, hosting is most strongly associated with housing space, household size, higher education, and exposure to Ukrainians at work. Notably, while non-Ukrainian hosts are much richer unconditionally (Supplementary Table S1), the income gradient largely disappears once housing capacity is controlled for (Supplementary Table S2, columns 3–4), suggesting that the apparent income effect operates primarily through the ability to provide space. Among households with a Ukrainian member, by contrast, hosting is much less tightly linked to the usual markers of socioeconomic advantage, although housing capacity still matters. Once close ties already exist, hosting can be supplied by a broader cross-section of households.

Because private hosting is a rare event, we complement the parametric model with a flexible prediction exercise. We estimate a weighted random forest and use permutation importance to rank predictors by their contribution to out-of-sample classification. The ranking in Figure 4 largely mirrors the regression results: housing-capacity measures and household structure dominate the prediction problem, with Ukrainian household ties and disposable income also ranking highly. The convergence between parametric and non-parametric exercises increases confidence that the main conclusions do not hinge on linearity assumptions.

Determinants of hosting duration

Most arrangements dissolve within a year, but those anchored in pre-existing Ukrainian household ties last somewhat longer. Among households without a Ukrainian member, 68% of arrangements dissolved within a year; among households with at least one Ukrainian member, the corresponding figure is 63%. The Kaplan-Meier curves (Figure 5) show that this durability gap opens within the first month and is sustained throughout the follow-up period. Among arrangements that dissolved, Ukrainian-connected households hosted for nearly three weeks longer on average (118 versus 99 days). Even so, durations are substantively long in both groups: typical arrangements lasted three to four months, and roughly a third of households were still hosting a full year after they began. Hosting was therefore not a brief act of hospitality but a sustained, months-long cohabitation with newly arrived refugees, reinforcing the high-cost character of the

Figure 4: Variable importance in random-forest prediction of hosting.

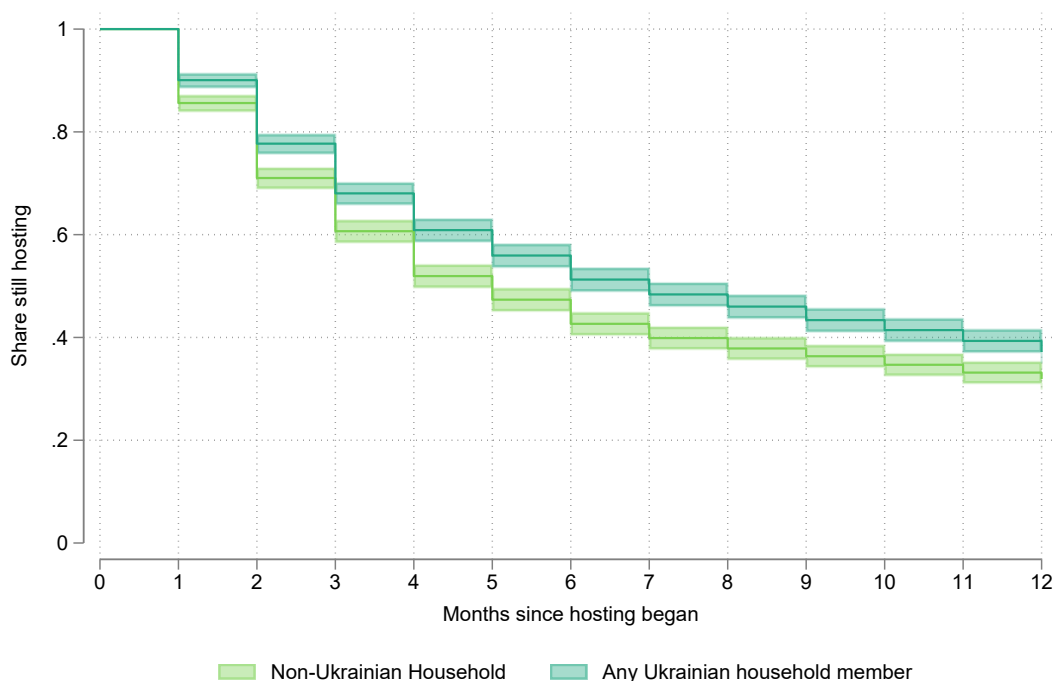


Note: Permutation-based variable importance from a weighted random forest that predicts whether a household privately hosted Ukrainian refugees. Higher values indicate a larger loss in predictive accuracy when the corresponding variable is permuted. The exercise is descriptive and intended to rank predictors rather than to identify causal effects.

behaviour.

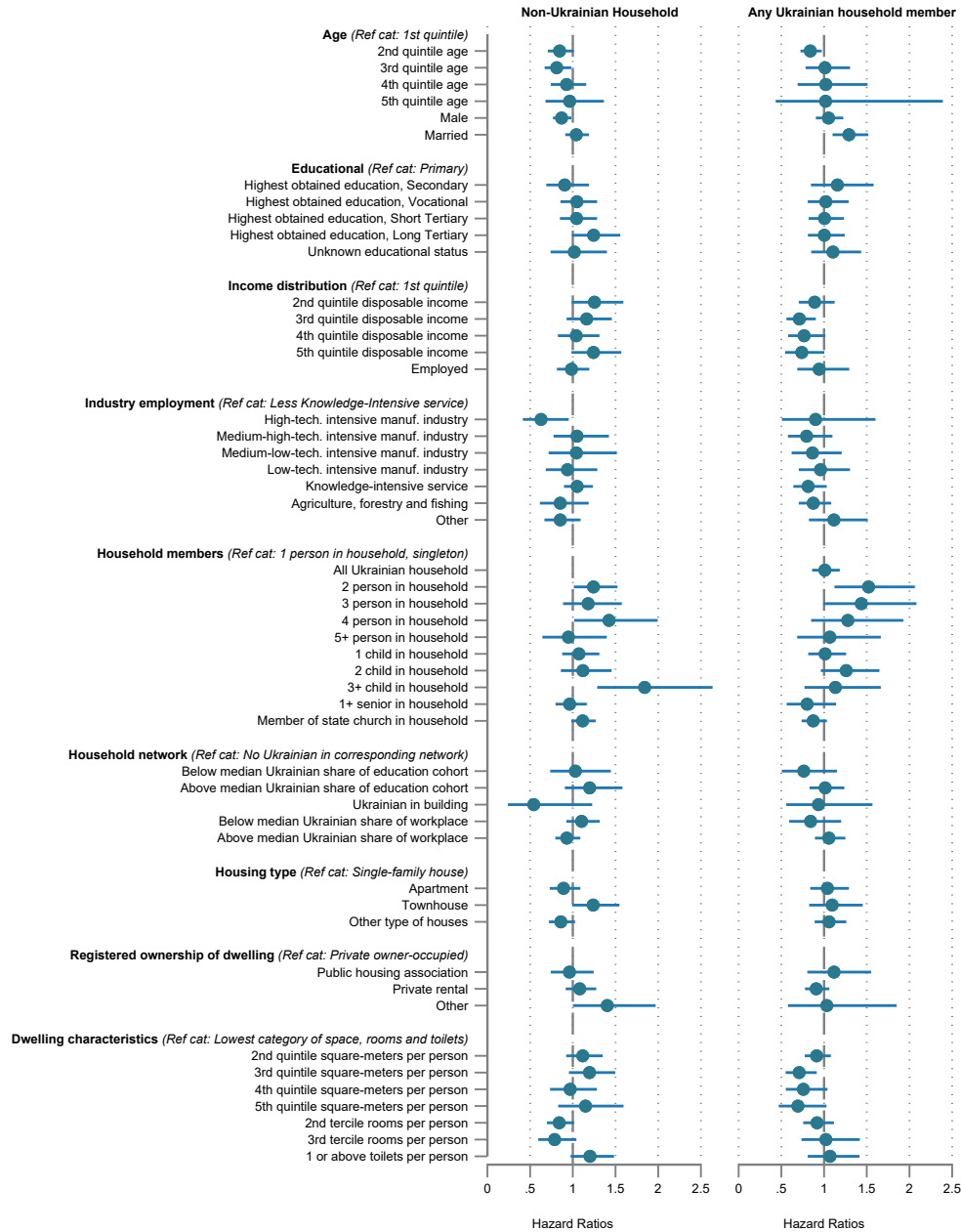
Cox proportional-hazards estimates (Figure 6; Supplementary Table S3) extend the descriptive picture by asking which household characteristics are associated with the duration of arrangements that did form. Several patterns echo the selection results. Among households with pre-existing Ukrainian ties, more space per person and higher disposable income are associated with longer arrangements—mirroring the same capacity gradients that predict who steps forward in the first place. Among hosts without a Ukrainian household member, who already constitute a positively selected, resource-rich subset of the population, these income and space gradients are largely absent, although hosts in less crowded dwellings (higher rooms-per-person) still show somewhat longer arrangements. The broader two-margin logic that organises selection also appears in durability: where ties are absent, capacity seems to operate mainly at the entry margin; where ties are present, it remains associated with the life of the arrangement. Other patterns are more subtle—hosts in lower-middle age quintiles and in higher-density municipalities show modestly longer arrangements, while townhouses and unconventional ownership categories are associated with shorter durations.

Figure 5: Kaplan-Meier survival curves of duration of private hosting



Note: The figure plots Kaplan-Meier survival curves for the duration of private hosting arrangements. The vertical axis shows the share of hosting arrangements still active at each point in time. The sample covers all hosting arrangements beginning after 24 February 2022, followed for up to 12 months (360 days); arrangements still active at 360 days are right-censored. Curves are estimated separately for households with and without a Ukrainian member at end-2021. Shaded bands are pointwise 95% confidence intervals.

Figure 6: Cox proportional-hazards estimates of hosting duration



Note: The figure plots hazard ratios from Cox proportional hazard models of hosting duration, estimated separately for households without a pre-existing Ukrainian household member (left panel) and households with at least one Ukrainian member at end-2021 (right panel). Horizontal bars are 95% confidence intervals. The vertical line marks a hazard ratio of 1 (no effect on exit probability). A hazard ratio below 1 corresponds to a lower instantaneous probability of ending the hosting arrangement (longer expected duration); a ratio above 1 corresponds to a higher probability of ending. Covariates and reference categories correspond to those in Supplementary Table S3.

Discussion

These results speak to long-running questions about the nature of costly prosocial behaviour. Theories of altruism have emphasised warm-glow motivations (Andreoni 1990), kin and group selection (Hamilton 1964; Fehr and Fischbacher 2003), and reputational incentives (Nowak and Sigmund 2005). Most empirical evidence comes from low-cost donations, hypothetical scenarios, or laboratory stakes; sustained material helping has been much harder to observe at scale. The Ukrainian crisis provided that rarer setting, and our population-level data indicate that helping under high stakes is neither randomly distributed nor a uniform expression of goodwill. Instead, it is associated with two concrete conditions: physical capacity to absorb the cost, and a path along which the helper is connected to the helped.

Three broader implications follow. First, capacity and proximity are not simply additive in these data; the patterns are consistent with partial substitution at important margins. Where social proximity is high, costly helping extends well beyond the most advantaged households. Where it is absent, the same act becomes a near-luxury good, supplied disproportionately by households with slack resources. Importantly, the absence of a pre-existing household tie need not imply complete strangerhood; it marks a broader set of socially more distant arrangements, many of which likely involved hosts and refugees who were not previously connected through the household. This pattern is also consistent with evidence that shared identity and perceived closeness can widen helping toward socially distant others (Levine et al. 2005; Hellmann, Fiedler, and Glöckner 2021). It also resonates with a long line of work on weak ties as bridges into otherwise unreachable populations (Granovetter 1973; Burt 2004), and it sharpens the prediction: the marginal supply of costly altruism is jointly bounded by the distribution of slack resources and by the network density connecting potential helpers to those in need. Population-scale civic capacity in crises depends on both, and may be more elastic where they overlap.

Second, what is striking is what does *not* predict hosting. Once room and ties are accounted for, broad markers of generic socioeconomic advantage add relatively little. The behaviour does not appear to track familiar indicators of "civic-mindedness" or community context as closely as one might expect; it tracks the specific structural conditions that make sustained, costly helping feasible. This refines accounts of civic capacity (Putnam 2000; Ostrom 1996) that have tended to emphasise diffuse social capital, and it extends work on volunteering that emphasises the role of capital endowments and embeddedness in enabling participation (Wilson and Musick 1997). The relevant capital here is concrete and dyadic: a spare room, and a thread of connection.

Third, ties are associated not only with entry into hosting but also with longer-lasting arrangements. Arrangements anchored in pre-existing Ukrainian connections are markedly more durable. Costly cooperation that relies on socially distant matches or weak ties may scale rapidly in the first weeks of a crisis, but its persistence over months appears to depend on the same proximity that enabled its initiation. This emphasis on persistence resonates with recent panel evidence from Poland showing that closeness and helping norms predicted continued assistance to Ukrainian refugees over time (Kossowska et al. 2023), as well as earlier work showing that sustained helping depends on more than initial willingness to help (Omoto and Snyder 1995). This has implications for how civic provision interacts with public reception systems (Feischmidt, Pries, and Cantat 2019): the substitutability of private for public capacity may be greatest in the early phase of a shock and erodes as informal arrangements unwind, particularly outside tied networks.

The analysis has limits. Administrative data are powerful for measuring realised hosting and the resources households bring to it, but they do not directly observe motivations; we therefore infer mechanisms from revealed selection patterns rather than from attitudes or stated preferences. Denmark is a relatively high-capacity institutional setting, and whether the same capacity-and-ties structure holds where state reception is weaker, network bridges to displaced populations are sparser, or housing markets are tighter remains an open empirical question. At the same time, a central strength of the design is that it identifies realised hosts in the full population rather than in a self-selected sample, making it possible to study costly prosocial behaviour directly rather than through proxies. Within those limits, the core finding is clear: at population scale, costly helping is associated with specific, observable combinations of material and social conditions—conditions that are unevenly distributed and that help explain when civic capacity is more likely to rise to meet a sudden shock.

Methods

Setting

Following Russia’s full-scale invasion of Ukraine on 24 February 2022, Denmark received approximately 25,000 Ukrainian refugees between March and June 2022 and roughly 40,000 by the end of 2023, a tenfold increase over typical baseline flows. Refugees were granted temporary protection under Denmark’s Special Act of 16 March 2022, which mirrored the EU Temporary Protection Directive and provided immediate residence and work rights. Public accommodation capacity did not expand quickly

enough to absorb the inflow during the peak weeks; private households accommodated a large share of arrivals during this period.

Identifying private hosts

We draw on Denmark’s linked population, migration, housing, and employment registers, which provide full-population information on residential histories, dwelling characteristics, household composition, education, and labour-market outcomes. Ukrainians admitted under the Special Act are identified from the migration and population registers. To focus on crisis-induced private hosting rather than co-residence with Ukrainians already settled in Denmark, we exclude Ukrainian citizens who were already resident in Denmark before 24 February 2022 from the refugee sample.

The central measurement challenge is that private hosting is not recorded as its own administrative status. Rather than observing hosts directly, we reconstruct hosting from building identifiers, address histories, and co-residence records.

We classify a hosting episode when a Ukrainian refugee is observed moving into a dwelling that was already occupied by at least one established resident, and that resident remains registered at the address throughout the refugee’s stay. This definition is designed to capture genuine within-household hosting rather than parallel arrival into empty or newly opened units. To avoid treating contemporaneous movers as hosts, we require potential hosts to be present at the address on 24 February 2022 and in the end-of-2021 population snapshot. The result is a population-wide measure of realised private hosts that can be linked back to pre-crisis household resources, housing capacity, and prior Ukrainian ties.

Further details on the accommodation classification are provided in Foged et al. (2026).

Sample and unit of analysis

For each privately hosting household, we identify a single focal individual—the household’s breadwinner—as the adult resident with the highest 2021 disposable income. In the rare case of exact income ties, priority is assigned deterministically by age and sex solely to ensure a unique focal individual. We apply the same procedure to all households in Denmark at the end of 2021, yielding a population-wide comparison group of non-hosting breadwinners. We then remove households where the oldest resident is below age 18 or that contain more than 20 members, to focus on settings in which

meaningful interaction between refugees and hosts is plausible. The final analysis sample contains approximately 2.67 million breadwinners, of whom roughly 0.15% (about 3,900) privately hosted Ukrainian refugees during the period studied. We include all private hosting arrangements we can record from 24 February 2022 to 31 December 2022.

Predictors

Covariates cover socio-demographic characteristics (age quintiles, sex, marital status, citizenship), education (five-category aggregation of highest completed education), and labour-market position in 2021 (employment status, industry mapped to a Eurostat high-technology taxonomy following Eurostat 2024, and disposable income quintiles). We also measure household composition using household size, number of children, presence of seniors, citizenship composition, and state-church membership. Housing conditions are captured using tenure, dwelling type, and several indicators of space and crowding, including square metres per person, rooms per person, and toilets per person. Finally, we include broad municipal context measures (log population density, right-wing party vote share, and sport-club participation per capita) and four proxies for prior exposure to Ukrainians: whether the household included a Ukrainian citizen at the end of 2021, whether another Ukrainian lived in the same building, whether the breadwinner belonged to an education-cohort network with Ukrainian presence, and whether the breadwinner's workplace had an above-median share of Ukrainian employees.

Statistical analysis

We proceed in three steps. First, we describe how hosts differ from the rest of the Danish population, both pooled and split by whether the household included a Ukrainian member at the end of 2021. Second, we estimate linear probability models for selection into private hosting, with municipality fixed effects to absorb local heterogeneity in housing markets, refugee inflows, and policy environments. Standard errors are robust to heteroskedasticity. To facilitate interpretation, we report coefficients rescaled by group-specific baseline hosting prevalence; rescaled coefficients can be read as proportional changes in the probability of hosting relative to baseline. Third, we estimate a weighted random forest to evaluate the predictive importance of the same covariates in a non-parametric setting, using permutation importance for ranking. All analyses were conducted in a secure research environment under a data-use agreement with Statistics Denmark.

Ethics and data protection

The study relied exclusively on secondary administrative register data and did not involve direct contact with participants, collection of new data, or use of biological materials. Under the Danish legal framework for register-based research, formal research-ethics committee approval and individual informed consent were not required for this study. Access to the data was granted through Statistics Denmark's secure research environment under a data-use agreement, and analyses were conducted in compliance with Danish data protection regulations. No individual-level information left the secure environment; only aggregate, non-identifying results were exported.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary accompanying this submission.

Data availability

The administrative register data used in this study are owned by Statistics Denmark and are not publicly available due to confidentiality requirements. Access can be obtained by qualified researchers affiliated with an authorised Danish research institution under a data-use agreement with Statistics Denmark.

Code availability

Replication code for all analyses will be made available to editors and reviewers during peer review and deposited in a public repository before publication. Code that can only be executed inside Statistics Denmark's secure research environment will be documented and made available to qualified researchers with approved data access.

Use of generative AI

Generative AI tools were used for language editing, proofreading, and manuscript preparation support. They were not used to generate data, conduct statistical analyses, or produce results. All AI-assisted edits were reviewed and approved by the authors.

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Competing interests

The authors declare no competing interests.

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Supplementary information

A Data details

Table S1: Summary statistics by hosting status and pre-existing Ukrainian ties.

	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1) Private Hosts	(2) Rest of Denmark	(3) Private Hosts	(4) Rest of Denmark	(5) Private Hosts	(6) Rest of Denmark
<i>Demographics</i>						
1st quintile age	0.318 (0.466)	0.210 (0.408)	0.162 (0.369)	0.210 (0.407)	0.481 (0.500)	0.513 (0.500)
2nd quintile age	0.369 (0.483)	0.190 (0.393)	0.356 (0.479)	0.190 (0.392)	0.383 (0.486)	0.329 (0.470)
3rd quintile age	0.194 (0.396)	0.199 (0.399)	0.289 (0.453)	0.199 (0.400)	0.096 (0.294)	0.092 (0.289)
4th quintile age	0.095 (0.294)	0.201 (0.400)	0.153 (0.361)	0.201 (0.401)	0.035 (0.183)	0.055 (0.228)
5th quintile age	0.023 (0.150)	0.199 (0.400)	0.039 (0.195)	0.200 (0.400)	0.006 (0.079)	0.011 (0.103)
Male	0.733 (0.442)	0.563 (0.496)	0.690 (0.463)	0.563 (0.496)	0.778 (0.416)	0.747 (0.435)
Married	0.610 (0.488)	0.380 (0.485)	0.495 (0.500)	0.380 (0.485)	0.729 (0.445)	0.663 (0.473)
<i>Socioeconomics</i>						
Highest obtained education, Primary	0.114 (0.318)	0.222 (0.415)	0.117 (0.321)	0.222 (0.416)	0.111 (0.315)	0.101 (0.302)
Highest obtained education, Secondary	0.059 (0.237)	0.082 (0.274)	0.071 (0.257)	0.082 (0.274)	0.047 (0.212)	0.058 (0.234)
Highest obtained education, Vocational	0.215 (0.411)	0.332 (0.471)	0.260 (0.439)	0.332 (0.471)	0.167 (0.373)	0.182 (0.386)
Highest obtained education, Short Tertiary	0.290 (0.454)	0.228 (0.420)	0.271 (0.444)	0.228 (0.420)	0.310 (0.462)	0.313 (0.464)
Highest obtained education, Long Tertiary	0.249 (0.432)	0.117 (0.321)	0.232 (0.422)	0.116 (0.321)	0.266 (0.442)	0.249 (0.432)
Unknown educational status	0.073 (0.261)	0.019 (0.138)	0.049 (0.217)	0.019 (0.137)	0.098 (0.298)	0.098 (0.297)
1st quintile disposable income	0.111 (0.314)	0.200 (0.400)	0.129 (0.336)	0.200 (0.400)	0.092 (0.289)	0.131 (0.337)
2nd quintile disposable income	0.193 (0.395)	0.200 (0.400)	0.111 (0.314)	0.200 (0.400)	0.279 (0.448)	0.278 (0.448)
3rd quintile disposable income	0.247 (0.431)	0.200 (0.400)	0.169 (0.375)	0.200 (0.400)	0.328 (0.469)	0.314 (0.464)
4th quintile disposable income	0.177 (0.382)	0.200 (0.400)	0.205 (0.404)	0.200 (0.400)	0.148 (0.355)	0.141 (0.348)
5th quintile disposable income	0.272 (0.445)	0.200 (0.400)	0.385 (0.487)	0.200 (0.400)	0.154 (0.361)	0.136 (0.343)
Employed	0.876 (0.329)	0.637 (0.481)	0.812 (0.391)	0.636 (0.481)	0.943 (0.231)	0.912 (0.283)

Table S1 continues next page

<i>Continuation of Table S1</i>						
	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1) Private Hosts	(2) Rest of Denmark	(3) Private Hosts	(4) Rest of Denmark	(5) Private Hosts	(6) Rest of Denmark
<i>Industrial employment</i>						
High-tech. intensive manuf. industry	0.017 (0.130)	0.012 (0.108)	0.023 (0.149)	0.012 (0.108)	0.012 (0.107)	0.006 (0.079)
Medium-high-tech. intensive manuf. industry	0.044 (0.204)	0.024 (0.153)	0.039 (0.193)	0.024 (0.153)	0.048 (0.215)	0.033 (0.179)
Medium-low-tech. intensive manuf. industry	0.033 (0.179)	0.021 (0.142)	0.024 (0.152)	0.021 (0.142)	0.043 (0.202)	0.035 (0.183)
Low-tech. intensive manuf. industry	0.042 (0.201)	0.024 (0.154)	0.036 (0.187)	0.024 (0.154)	0.048 (0.215)	0.042 (0.201)
Knowledge-intensive service	0.261 (0.439)	0.323 (0.468)	0.375 (0.484)	0.323 (0.468)	0.142 (0.350)	0.170 (0.375)
Less-knowledge-intensive service	0.163 (0.370)	0.172 (0.377)	0.201 (0.401)	0.172 (0.378)	0.124 (0.329)	0.136 (0.343)
Agriculture, forestry and fishing	0.254 (0.436)	0.008 (0.087)	0.038 (0.192)	0.007 (0.083)	0.479 (0.500)	0.449 (0.497)
Other	0.062 (0.240)	0.053 (0.225)	0.076 (0.265)	0.053 (0.225)	0.047 (0.211)	0.042 (0.200)
<i>Household composition</i>						
Any Ukrainian household member	0.490 (0.500)	0.002 (0.040)	0.000 (0.000)	0.000 (0.000)	1.000 (0.000)	1.000 (0.000)
All Ukrainian household	0.285 (0.451)	0.001 (0.029)	0.000 (0.000)	0.000 (0.000)	0.581 (0.494)	0.530 (0.499)
1 person in household	0.200 (0.400)	0.400 (0.490)	0.282 (0.450)	0.401 (0.490)	0.115 (0.319)	0.131 (0.338)
2 person in household	0.235 (0.424)	0.336 (0.472)	0.266 (0.442)	0.337 (0.473)	0.204 (0.403)	0.279 (0.448)
3 person in household	0.216 (0.411)	0.111 (0.314)	0.156 (0.363)	0.111 (0.314)	0.278 (0.448)	0.256 (0.437)
4 person in household	0.214 (0.410)	0.106 (0.308)	0.174 (0.379)	0.106 (0.308)	0.255 (0.436)	0.206 (0.405)
5+ person in household	0.135 (0.342)	0.046 (0.209)	0.123 (0.328)	0.046 (0.209)	0.148 (0.355)	0.128 (0.334)
1 child in household	0.217 (0.412)	0.106 (0.308)	0.160 (0.366)	0.106 (0.307)	0.276 (0.447)	0.259 (0.438)
2 child in household	0.202 (0.402)	0.104 (0.305)	0.168 (0.374)	0.104 (0.305)	0.238 (0.426)	0.188 (0.391)
3+ child in household	0.062 (0.241)	0.034 (0.182)	0.070 (0.255)	0.034 (0.182)	0.054 (0.225)	0.043 (0.204)
1+ senior in household	0.094 (0.292)	0.199 (0.399)	0.144 (0.352)	0.199 (0.399)	0.042 (0.200)	0.051 (0.219)
Member of state church in household	0.448 (0.497)	0.848 (0.359)	0.630 (0.483)	0.849 (0.358)	0.258 (0.437)	0.258 (0.438)
<i>Household network</i>						
No Ukrainian in education cohort	0.908 (0.290)	0.959 (0.197)	0.937 (0.243)	0.960 (0.197)	0.877 (0.329)	0.884 (0.320)
Below median Ukrainian share of education cohort	0.024 (0.155)	0.021 (0.142)	0.027 (0.161)	0.021 (0.142)	0.022 (0.147)	0.023 (0.149)
<i>Table S1 continues next page</i>						

<i>Continuation of Table S1</i>						
	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1) Private Hosts	(2) Rest of Denmark	(3) Private Hosts	(4) Rest of Denmark	(5) Private Hosts	(6) Rest of Denmark
Above median Ukrainian share of education cohort	0.068 (0.252)	0.020 (0.140)	0.036 (0.187)	0.020 (0.139)	0.101 (0.301)	0.093 (0.290)
No Ukrainian in building	0.989 (0.106)	0.995 (0.072)	0.992 (0.090)	0.995 (0.071)	0.985 (0.120)	0.973 (0.161)
Ukrainian in building	0.011 (0.106)	0.005 (0.072)	0.008 (0.090)	0.005 (0.071)	0.015 (0.120)	0.027 (0.161)
No Ukrainian in workplace	0.453 (0.498)	0.821 (0.383)	0.701 (0.458)	0.822 (0.383)	0.195 (0.396)	0.258 (0.438)
Below median Ukrainian share of workplace	0.082 (0.274)	0.090 (0.286)	0.120 (0.325)	0.090 (0.286)	0.043 (0.202)	0.041 (0.198)
Above median Ukrainian share of workplace	0.465 (0.499)	0.089 (0.285)	0.179 (0.384)	0.088 (0.284)	0.763 (0.425)	0.701 (0.458)
<i>Housing type</i>						
Apartment	0.211 (0.408)	0.378 (0.485)	0.234 (0.423)	0.378 (0.485)	0.188 (0.391)	0.302 (0.459)
Townhouse	0.062 (0.241)	0.151 (0.358)	0.073 (0.260)	0.151 (0.358)	0.051 (0.220)	0.082 (0.275)
Single-family house	0.488 (0.500)	0.394 (0.489)	0.519 (0.500)	0.394 (0.489)	0.456 (0.498)	0.363 (0.481)
Other type of houses	0.247 (0.431)	0.079 (0.270)	0.186 (0.389)	0.079 (0.270)	0.310 (0.462)	0.255 (0.436)
<i>Registered ownership of dwelling</i>						
Public housing association	0.060 (0.238)	0.192 (0.394)	0.078 (0.269)	0.192 (0.394)	0.041 (0.198)	0.085 (0.279)
Private owner-occupied	0.515 (0.500)	0.481 (0.500)	0.624 (0.485)	0.481 (0.500)	0.401 (0.490)	0.322 (0.467)
Private rental	0.351 (0.477)	0.250 (0.433)	0.221 (0.415)	0.249 (0.433)	0.486 (0.500)	0.494 (0.500)
Other	0.036 (0.186)	0.055 (0.228)	0.047 (0.212)	0.055 (0.228)	0.025 (0.155)	0.034 (0.180)
<i>Dwelling characteristics</i>						
1st quintile square-meters per person	0.268 (0.443)	0.201 (0.401)	0.191 (0.393)	0.201 (0.400)	0.347 (0.476)	0.439 (0.496)
2nd quintile square-meters per person	0.236 (0.425)	0.201 (0.401)	0.192 (0.394)	0.201 (0.400)	0.283 (0.450)	0.246 (0.431)
3rd quintile square-meters per person	0.160 (0.366)	0.201 (0.401)	0.180 (0.384)	0.201 (0.401)	0.139 (0.346)	0.138 (0.345)
4th quintile square-meters per person	0.128 (0.334)	0.200 (0.400)	0.152 (0.359)	0.200 (0.400)	0.104 (0.305)	0.090 (0.287)
5th quintile square-meters per person	0.208 (0.406)	0.197 (0.398)	0.286 (0.452)	0.198 (0.398)	0.128 (0.334)	0.086 (0.281)
1st tercile rooms per person	0.464 (0.499)	0.371 (0.483)	0.365 (0.482)	0.370 (0.483)	0.565 (0.496)	0.639 (0.480)
2nd tercile rooms per person	0.286 (0.452)	0.352 (0.477)	0.292 (0.455)	0.352 (0.477)	0.280 (0.449)	0.241 (0.428)
3rd tercile rooms per person	0.250 (0.433)	0.278 (0.448)	0.343 (0.475)	0.278 (0.448)	0.155 (0.362)	0.120 (0.325)
<i>Table S1 continues next page</i>						

<i>Continuation of Table S1</i>						
	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1) Private Hosts	(2) Rest of Denmark	(3) Private Hosts	(4) Rest of Denmark	(5) Private Hosts	(6) Rest of Denmark
Below 1 toilets per person	0.683 (0.465)	0.477 (0.499)	0.560 (0.497)	0.476 (0.499)	0.810 (0.392)	0.812 (0.391)
1 or above toilets per person	0.317 (0.465)	0.523 (0.499)	0.440 (0.497)	0.524 (0.499)	0.190 (0.392)	0.188 (0.391)
<i>Municipality characteristics</i>						
Log-population density (population/sq km)	5.207 (1.594)	5.740 (1.643)	5.671 (1.684)	5.741 (1.643)	4.722 (1.333)	4.984 (1.517)
Right-wing party share, General, 2019	0.136 (0.042)	0.128 (0.042)	0.131 (0.044)	0.128 (0.042)	0.142 (0.038)	0.137 (0.040)
Sportclub members per person	0.449 (0.081)	0.437 (0.086)	0.441 (0.082)	0.437 (0.086)	0.456 (0.080)	0.454 (0.079)
<i>N</i>	3883	2668898	1981	2664640	1902	4258

Notes: The table reports means for breadwinners in private-hosting households and in the rest of Denmark. Columns (3)–(6) split the sample by whether the household included at least one Ukrainian citizen at the end of 2021, before the large-scale inflow following Russia's invasion.

B Predictors of selection into private hosting

Table S2: Predictors of selection into private hosting.

	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Age (Ref cat: 1st quintile)</i>						
2nd quintile age	0.069*** (0.010)	0.068*** (0.010)	0.057*** (0.007)	0.056*** (0.007)	3.354** (1.538)	3.117** (1.543)
3rd quintile age	0.025*** (0.008)	0.025*** (0.008)	0.024*** (0.006)	0.023*** (0.006)	2.663 (2.456)	2.923 (2.458)
4th quintile age	-0.015* (0.009)	-0.016* (0.009)	-0.014* (0.007)	-0.015** (0.007)	-6.424** (3.199)	-4.822 (3.211)
5th quintile age	-0.035*** (0.009)	-0.036*** (0.009)	-0.039*** (0.007)	-0.040*** (0.007)	-0.820 (5.791)	1.221 (5.749)
Male	0.018*** (0.004)	0.018*** (0.004)	0.016*** (0.004)	0.016*** (0.004)	0.602 (1.433)	0.693 (1.446)
Married	0.009 (0.008)	0.009 (0.008)	0.003 (0.006)	0.003 (0.006)	3.688** (1.530)	3.307** (1.537)
<i>Educational (Ref cat: Primary)</i>						
Highest obtained education, Secondary	0.022** (0.009)	0.022** (0.009)	0.020*** (0.007)	0.020*** (0.007)	-5.149* (2.942)	-5.881** (2.951)
Highest obtained education, Vocational	-0.010** (0.005)	-0.010** (0.005)	0.001 (0.004)	0.001 (0.004)	-5.967*** (2.307)	-6.278*** (2.295)
Highest obtained education, Short Tertiary	0.021*** (0.004)	0.021*** (0.004)	0.029*** (0.004)	0.029*** (0.004)	-4.600** (2.307)	-5.064** (2.295)

Table S2 continues next page

<i>Continuation of Table S2</i>						
	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1)	(2)	(3)	(4)	(5)	(6)
	(0.007)	(0.007)	(0.005)	(0.005)	(2.128)	(2.123)
Highest obtained education, Long Tertiary	0.063*** (0.011)	0.062*** (0.012)	0.066*** (0.008)	0.065*** (0.008)	-3.182 (2.244)	-3.823* (2.238)
Unknown educational status	0.094*** (0.028)	0.094*** (0.028)	0.101*** (0.019)	0.101*** (0.019)	-2.599 (2.702)	-3.491 (2.716)
<i>Income distribution (Ref cat: 1st quintile)</i>						
2nd quintile disposable income	-0.010 (0.006)	-0.011* (0.006)	-0.011** (0.004)	-0.012** (0.005)	1.576 (2.178)	1.366 (2.184)
3rd quintile disposable income	-0.024*** (0.008)	-0.026*** (0.008)	-0.025*** (0.006)	-0.026*** (0.006)	0.914 (2.221)	0.811 (2.246)
4th quintile disposable income	-0.045*** (0.009)	-0.047*** (0.009)	-0.038*** (0.007)	-0.039*** (0.007)	2.148 (2.620)	1.914 (2.632)
5th quintile disposable income	-0.017 (0.011)	-0.022** (0.011)	-0.003 (0.009)	-0.006 (0.009)	5.024* (2.791)	4.485 (2.824)
Employed	0.012 (0.008)	0.012 (0.008)	0.010 (0.007)	0.010 (0.007)	4.803* (2.682)	4.557* (2.694)
<i>Industry employment (Ref cat: Less Knowledge-Intensive service)</i>						
High-tech. intensive manuf. industry	0.044* (0.025)	0.045* (0.025)	0.020 (0.022)	0.021 (0.022)	16.441** (7.408)	16.190** (7.514)
Medium-high-tech. intensive manuf. industry	0.037* (0.019)	0.035* (0.019)	0.009 (0.015)	0.007 (0.015)	6.621* (3.600)	8.135** (3.562)
Medium-low-tech. intensive manuf. industry	-0.002 (0.019)	-0.002 (0.019)	-0.015 (0.014)	-0.015 (0.014)	2.922 (3.577)	2.194 (3.592)
Low-tech. intensive manuf. industry	0.010 (0.019)	0.009 (0.019)	0.004 (0.014)	0.003 (0.014)	0.148 (3.289)	-0.311 (3.300)
Knowledge-intensive service	-0.011* (0.007)	-0.010 (0.007)	-0.010* (0.006)	-0.009* (0.006)	0.088 (2.185)	-0.955 (2.188)
Agriculture, forestry and fishing	0.263*** (0.093)	0.263*** (0.093)	0.263*** (0.046)	0.263*** (0.046)	-1.970 (2.237)	-2.011 (2.249)
Other	0.014 (0.012)	0.015 (0.012)	0.011 (0.010)	0.011 (0.010)	1.628 (3.267)	0.815 (3.250)
<i>Household members (Ref cat: 1 person in household, singleton)</i>						
Any Ukrainian household member	28.163*** (0.852)	28.160*** (0.852)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
All Ukrainian household	4.231*** (1.178)	4.233*** (1.178)	0.000 (.)	0.000 (.)	5.923*** (1.673)	5.922*** (1.674)
2 person in household	0.056*** (0.012)	0.055*** (0.012)	0.053*** (0.010)	0.052*** (0.010)	3.085 (3.040)	3.659 (3.059)
3 person in household	0.140*** (0.018)	0.137*** (0.018)	0.104*** (0.015)	0.102*** (0.015)	10.100*** (3.728)	10.088*** (3.752)
4 person in household	0.203*** (0.027)	0.200*** (0.027)	0.148*** (0.021)	0.146*** (0.021)	13.511*** (4.181)	13.851*** (4.202)

Table S2 continues next page

<i>Continuation of Table S2</i>						
	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1)	(2)	(3)	(4)	(5)	(6)
5+ person in household	0.363*** (0.044)	0.359*** (0.044)	0.278*** (0.032)	0.275*** (0.032)	14.992*** (4.451)	15.363*** (4.492)
1 child in household	-0.001 (0.013)	-0.001 (0.013)	-0.007 (0.010)	-0.007 (0.010)	0.241 (2.101)	0.903 (2.101)
2 child in household	-0.036* (0.020)	-0.035* (0.020)	-0.045*** (0.015)	-0.045*** (0.015)	1.669 (2.787)	2.245 (2.805)
3+ child in household	-0.149*** (0.039)	-0.147*** (0.039)	-0.130*** (0.029)	-0.129*** (0.029)	1.560 (4.004)	2.200 (4.028)
1+ senior in household	-0.012 (0.009)	-0.012 (0.009)	-0.012 (0.008)	-0.012 (0.008)	-0.755 (3.185)	-1.579 (3.186)
Member of state church in household	-0.117*** (0.008)	-0.115*** (0.008)	-0.121*** (0.007)	-0.120*** (0.007)	0.887 (1.669)	0.691 (1.689)
<i>Household network (Ref cat: No Ukrainian in corresponding network)</i>						
Below median Ukrainian share of education cohort	-0.015 (0.017)	-0.016 (0.017)	-0.008 (0.014)	-0.009 (0.014)	-0.766 (3.960)	-0.716 (3.988)
Above median Ukrainian share of education cohort	0.015 (0.027)	0.012 (0.027)	0.014 (0.017)	0.011 (0.017)	-3.554* (2.152)	-3.757* (2.164)
Ukrainian in building	-0.059 (0.046)	-0.062 (0.046)	0.048 (0.029)	0.045 (0.029)	-3.132 (3.571)	-2.134 (3.658)
Below median Ukrainian share of workplace	0.002 (0.008)	0.002 (0.008)	-0.002 (0.007)	-0.001 (0.007)	4.080 (3.124)	4.229 (3.135)
Above median Ukrainian share of workplace	0.068*** (0.011)	0.065*** (0.011)	0.044*** (0.008)	0.042*** (0.008)	3.578** (1.593)	3.882** (1.610)
<i>Housing type (Ref cat: Single-family house)</i>						
Apartment	-0.031*** (0.008)	-0.034*** (0.008)	-0.016** (0.006)	-0.018*** (0.006)	-5.028** (1.962)	-4.792** (1.981)
Townhouse	-0.028*** (0.007)	-0.028*** (0.007)	-0.014*** (0.006)	-0.014** (0.006)	-8.089*** (2.367)	-7.314*** (2.379)
Other type of houses	0.130*** (0.016)	0.131*** (0.016)	0.091*** (0.012)	0.092*** (0.012)	-0.791 (1.828)	-0.330 (1.836)
<i>Registered ownership of dwelling (Ref cat: Private owner-occupied)</i>						
Public housing association	0.015** (0.008)	0.015* (0.008)	0.003 (0.006)	0.003 (0.006)	-0.653 (2.474)	-0.420 (2.481)
Private rental	0.031*** (0.008)	0.031*** (0.008)	0.017*** (0.006)	0.017*** (0.006)	0.891 (1.567)	0.638 (1.568)
Other	0.040*** (0.012)	0.039*** (0.012)	0.032*** (0.010)	0.031*** (0.010)	0.419 (4.504)	1.180 (4.385)
<i>Dwelling characteristics (Ref cat: 1st quintile square-meters per person)</i>						
2nd quintile square-meters per person	0.092***	0.092***	0.048***	0.048***	6.784***	6.297***
<i>Table S2 continues next page</i>						

<i>Continuation of Table S2</i>						
	All		No Ukrainian HH member		Any Ukrainian HH member	
	(1)	(2)	(3)	(4)	(5)	(6)
	(0.009)	(0.009)	(0.006)	(0.006)	(1.667)	(1.669)
3rd quintile square-meters per person	0.130*** (0.012)	0.129*** (0.012)	0.085*** (0.009)	0.084*** (0.009)	5.077** (2.546)	4.771* (2.550)
4th quintile square-meters per person	0.143*** (0.013)	0.144*** (0.014)	0.090*** (0.010)	0.090*** (0.010)	9.343*** (3.229)	8.437*** (3.256)
5th quintile square-meters per person	0.194*** (0.016)	0.194*** (0.016)	0.132*** (0.012)	0.131*** (0.012)	17.610*** (4.115)	16.861*** (4.114)
<i>Dwelling characteristics (Ref cat: 1st tercile rooms per person)</i>						
2nd tercile rooms per person	0.018** (0.008)	0.018** (0.008)	0.006 (0.006)	0.006 (0.006)	3.602* (2.067)	3.917* (2.075)
3rd tercile rooms per person	0.040*** (0.011)	0.039*** (0.011)	0.031*** (0.008)	0.031*** (0.008)	2.523 (3.537)	3.516 (3.526)
<i>Dwelling characteristics (Ref cat: Below 1 toilets per person)</i>						
1 or above toilets per person	0.012 (0.009)	0.011 (0.009)	0.006 (0.007)	0.005 (0.007)	2.234 (2.735)	2.364 (2.741)
<i>Municipality characteristics</i>						
Log-population density (population/sq km)	0.007*** (0.002)	0.000 (.)	0.005*** (0.002)	0.000 (.)	-0.807 (0.679)	0.000 (.)
Right-wing party share, General, 2019	0.242*** (0.080)	0.000 (.)	0.239*** (0.062)	0.000 (.)	8.453 (20.893)	0.000 (.)
Sportclub members per person	0.027 (0.025)	0.000 (.)	0.043** (0.020)	0.000 (.)	-9.543 (8.679)	0.000 (.)
Constant	-0.140*** (0.033)	-0.053*** (0.019)	-0.065** (0.026)	0.018 (0.015)	12.238 (9.682)	5.654 (4.813)
Mean of dep. var. (Percent)	0.145	0.145	0.074	0.074	30.877	30.887
Observations	2,672,781	2,672,781	2,666,621	2,666,621	6,160	6,158
R ²	0.152	0.152	0.001	0.001	0.050	0.076
Municipality FE		✓		✓		✓

Notes: Linear probability models for selection into private hosting. Columns (1)–(2) use the full population, columns (3)–(4) restrict the sample to households without a Ukrainian member at the end of 2021, and columns (5)–(6) restrict the sample to households with a Ukrainian member. Municipality fixed effects are included where indicated. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Predictors of duration of hosting

Table S3: Duration modelling: Cox proportional hazard estimates.

	All	No Ukrainian HH member	Any Ukrainian HH member
	Hazard Ratios (1)	Hazard Ratios (2)	Hazard Ratios (3)
<i>Age (Ref cat: 1st quintile)</i>			
2nd quintile age	0.836*** [0.748,0.935]	0.846* [0.707,1.011]	0.839** [0.724,0.973]
3rd quintile age	0.865** [0.751,0.998]	0.812** [0.671,0.982]	1.011 [0.784,1.304]
4th quintile age	0.962 [0.804,1.151]	0.927 [0.742,1.157]	1.021 [0.690,1.510]
5th quintile age	1.039 [0.765,1.410]	0.963 [0.681,1.364]	1.019 [0.434,2.391]
Male	0.941 [0.857,1.034]	0.869** [0.768,0.983]	1.053 [0.905,1.225]
Married	1.127** [1.020,1.245]	1.041 [0.912,1.188]	1.292*** [1.100,1.517]
<i>Educational (Ref cat: Primary)</i>			
Highest obtained education, Secondary	1.005 [0.820,1.231]	0.905 [0.689,1.189]	1.157 [0.847,1.581]
Highest obtained education, Vocational	1.048 [0.903,1.218]	1.048 [0.856,1.284]	1.021 [0.809,1.287]
Highest obtained education, Short Tertiary	1.032 [0.894,1.192]	1.044 [0.849,1.284]	1.006 [0.820,1.234]
Highest obtained education, Long Tertiary	1.145* [0.984,1.333]	1.244* [0.994,1.555]	1.003 [0.811,1.242]
Unknown educational status	1.117 [0.919,1.359]	1.018 [0.740,1.399]	1.104 [0.850,1.434]
<i>Income distribution (Ref cat: 1st quintile)</i>			
2nd quintile disposable income	1.120 [0.953,1.317]	1.254* [0.988,1.591]	0.890 [0.705,1.124]
3rd quintile disposable income	0.960 [0.817,1.127]	1.162 [0.927,1.456]	0.711*** [0.560,0.903]
4th quintile disposable income	0.943 [0.791,1.124]	1.040 [0.825,1.312]	0.767* [0.579,1.015]
5th quintile disposable income	1.075 [0.899,1.286]	1.242* [0.984,1.568]	0.739* [0.545,1.002]
Employed	1.015 [0.865,1.192]	0.985 [0.813,1.193]	0.943 [0.686,1.295]
<i>Industry employment (Ref cat: Less Knowledge-Intensive service)</i>			
High-tech. intensive manuf. industry	0.729* [0.521,1.020]	0.628** [0.415,0.952]	0.900 [0.506,1.602]
Medium-high-tech. intensive manuf. industry	0.922 [0.741,1.148]	1.049 [0.774,1.423]	0.796 [0.577,1.098]
Medium-low-tech. intensive manuf. industry	0.956 [0.750,1.220]	1.042 [0.717,1.515]	0.865 [0.620,1.206]
Low-tech. intensive manuf. industry	0.932	0.938	0.959

Table S3 continues next page

<i>Continuation of Table S3</i>			
	All	No Ukrainian HH member	Any Ukrainian HH member
	Hazard Ratios (1)	Hazard Ratios (2)	Hazard Ratios (3)
	[0.750,1.158]	[0.683,1.286]	[0.706,1.303]
Knowledge-intensive service	1.014 [0.891,1.154]	1.051 [0.896,1.234]	0.813* [0.641,1.032]
Agriculture, forestry and fishing	0.911 [0.776,1.071]	0.853 [0.614,1.185]	0.873 [0.704,1.083]
Other	0.934 [0.772,1.128]	0.853 [0.668,1.090]	1.116 [0.823,1.512]
<i>Household members (Ref cat: 1 person in household, singleton)</i>			
Any Ukrainian household member	0.719*** [0.638,0.810]	1.000 [1.000,1.000]	1.000 [1.000,1.000]
All Ukrainian household	1.039 [0.903,1.195]	1.000 [1.000,1.000]	1.009 [0.860,1.184]
2 person in household	1.351*** [1.147,1.591]	1.242** [1.013,1.522]	1.522*** [1.122,2.063]
3 person in household	1.302** [1.044,1.624]	1.180 [0.885,1.573]	1.436* [0.991,2.083]
4 person in household	1.377** [1.072,1.768]	1.424** [1.017,1.994]	1.279 [0.849,1.928]
5+ person in household	1.023 [0.774,1.351]	0.948 [0.644,1.397]	1.068 [0.685,1.665]
1 child in household	1.076 [0.932,1.243]	1.071 [0.876,1.311]	1.012 [0.815,1.258]
2 child in household	1.211** [1.007,1.456]	1.118 [0.860,1.454]	1.260* [0.963,1.648]
3+ child in household	1.590*** [1.235,2.048]	1.841*** [1.286,2.637]	1.133 [0.771,1.664]
1+ senior in household	0.907 [0.772,1.065]	0.963 [0.797,1.163]	0.801 [0.563,1.142]
Member of state church in household	1.029 [0.933,1.136]	1.116* [0.982,1.268]	0.873 [0.737,1.033]
<i>Household network (Ref cat: No Ukrainian in corresponding network)</i>			
Below median Ukrainian share of education cohort	0.942 [0.728,1.220]	1.031 [0.735,1.445]	0.763 [0.505,1.151]
Above median Ukrainian share of education cohort	1.049 [0.893,1.231]	1.197 [0.907,1.581]	1.013 [0.830,1.237]
Ukrainian in building	0.774 [0.503,1.190]	0.542 [0.240,1.227]	0.935 [0.558,1.567]
Below median Ukrainian share of workplace	1.056 [0.903,1.234]	1.103 [0.926,1.315]	0.842 [0.591,1.200]
Above median Ukrainian share of workplace	0.982 [0.880,1.095]	0.930 [0.795,1.088]	1.056 [0.893,1.250]
<i>Housing type (Ref cat: Single-family house)</i>			
<i>Table S3 continues next page</i>			

<i>Continuation of Table S3</i>			
	All	No Ukrainian HH member	Any Ukrainian HH member
	Hazard Ratios (1)	Hazard Ratios (2)	Hazard Ratios (3)
Apartment	0.967 [0.838,1.117]	0.891 [0.730,1.088]	1.040 [0.838,1.291]
Townhouse	1.201** [1.011,1.427]	1.238* [0.992,1.546]	1.095 [0.826,1.452]
Other type of houses	0.907 [0.805,1.022]	0.860* [0.720,1.026]	1.059 [0.890,1.260]
<i>Registered ownership of dwelling (Ref cat: Private owner-occupied)</i>			
Public housing association	0.965 [0.790,1.179]	0.961 [0.742,1.245]	1.117 [0.804,1.551]
Private rental	1.006 [0.900,1.124]	1.081 [0.917,1.274]	0.908 [0.777,1.061]
Other	1.327* [0.997,1.767]	1.404** [1.001,1.969]	1.033 [0.577,1.850]
<i>Dwelling characteristics (Ref cat: 1st quintile square-meters per person)</i>			
2nd quintile square-meters per person	1.025 [0.907,1.158]	1.116 [0.923,1.350]	0.914 [0.774,1.079]
3rd quintile square-meters per person	0.997 [0.847,1.174]	1.195 [0.955,1.497]	0.709*** [0.549,0.915]
4th quintile square-meters per person	0.898 [0.732,1.102]	0.969 [0.733,1.280]	0.758* [0.552,1.041]
5th quintile square-meters per person	0.992 [0.776,1.267]	1.149 [0.829,1.594]	0.694* [0.469,1.028]
<i>Dwelling characteristics (Ref cat: 1st tercile rooms per person)</i>			
2nd tercile rooms per person	0.877* [0.767,1.002]	0.841* [0.697,1.015]	0.918 [0.756,1.116]
3rd tercile rooms per person	0.879 [0.713,1.084]	0.787* [0.595,1.042]	1.022 [0.736,1.419]
<i>Dwelling characteristics (Ref cat: Below 1 toilets per person)</i>			
1 or above toilets per person	1.124 [0.954,1.325]	1.201* [0.974,1.482]	1.070 [0.809,1.416]
<i>Municipality characteristics</i>			
Log-population density (population/sq km)	0.930*** [0.889,0.974]	0.940** [0.885,0.999]	0.962 [0.893,1.036]
Right-wing party share, General, 2019	0.308 [0.075,1.271]	0.725 [0.097,5.413]	0.205 [0.026,1.593]
Sportclub members per person	1.054 [0.606,1.833]	0.807 [0.369,1.764]	1.409 [0.623,3.189]
Observations	3,883	1,981	1,902
<i>Table S3 continues next page</i>			

<i>Continuation of Table S3</i>			
	All	No Ukrainian HH member	Any Ukrainian HH member
	Hazard Ratios (1)	Hazard Ratios (2)	Hazard Ratios (3)
Dissolved hosting arrangements within 360 days	2,544	1,349	1,195
Share dissolved within 360 days	0.655	0.681	0.628
Mean duration (dissolved)	107.8	98.9	117.8
Mean duration (all, censored at 360 days)	194.8	182.2	207.8
Harrell's C	0.585	0.600	0.592

Notes: Cox proportional hazard model estimates of hosting duration. The outcome is time from the start of a hosting arrangement until the arrangement ends; arrangements still active at 360 days are right-censored at that threshold. Columns (1)–(3) report estimates for the full sample, households without a Ukrainian member at end-2021, and households with at least one Ukrainian member, respectively. Hazard ratios are reported throughout; values above 1 indicate a higher instantaneous probability of exit (shorter expected duration) and values below 1 indicate a lower probability (longer expected duration). Ukrainian citizenship and Ukrainian household membership variables are collinear with the sample restriction in the relevant columns and are omitted by construction. Harrell's C measures concordance between predicted and observed ordering of event times. Ninety-five percent confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.